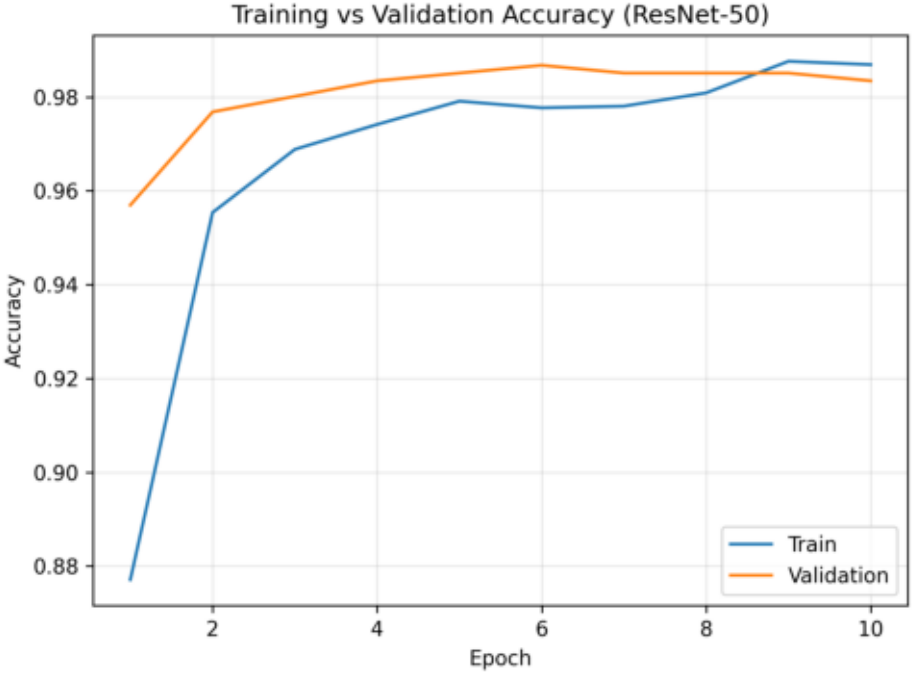
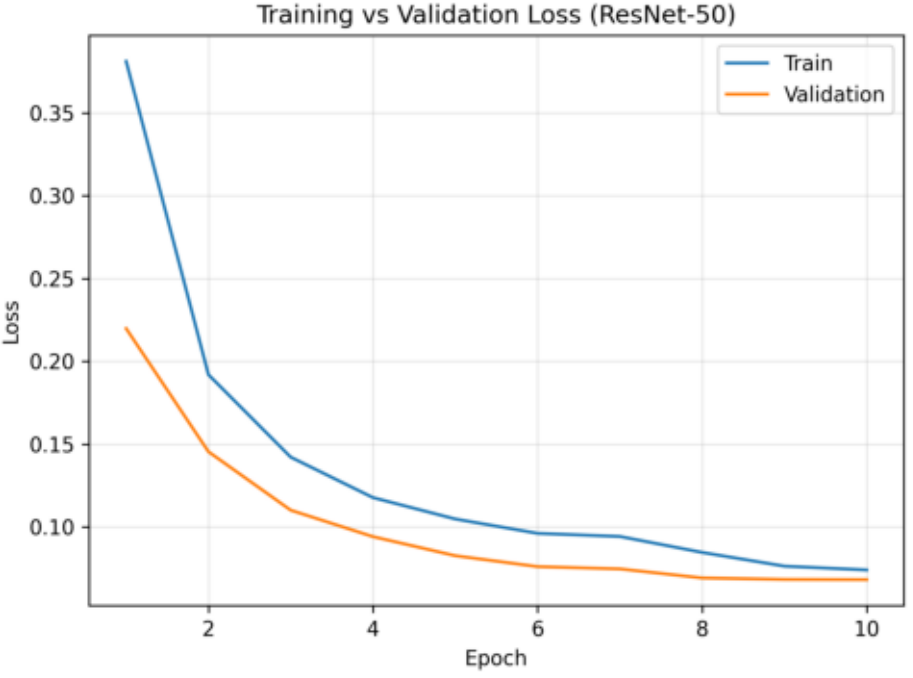
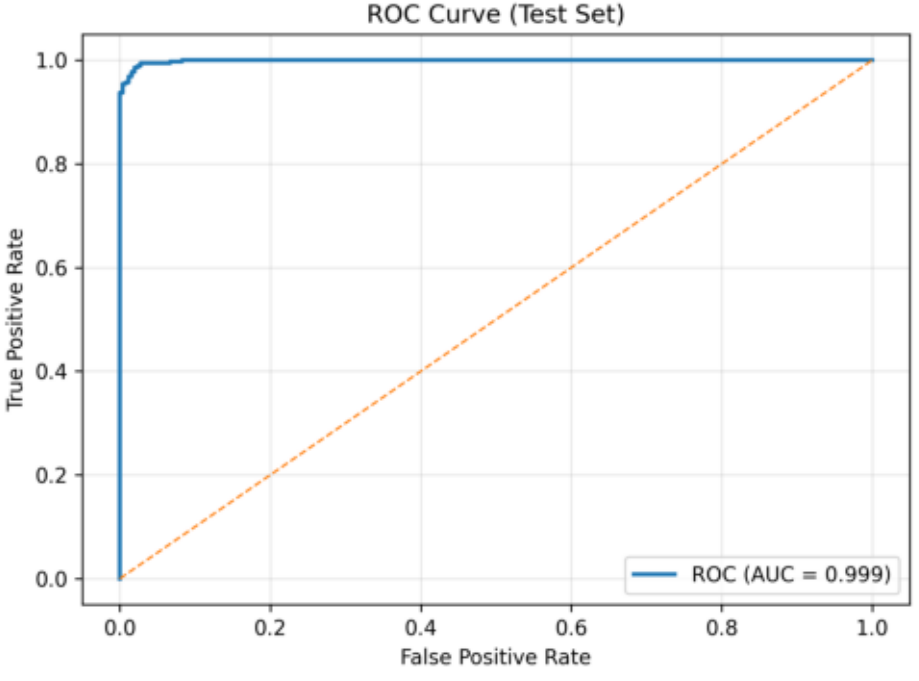
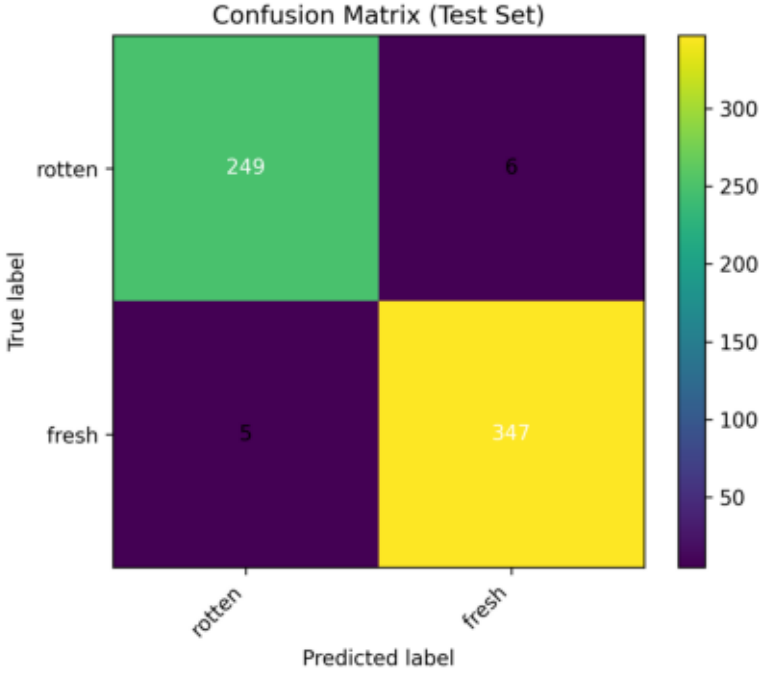
ResNet-50 Apple Freshness Classification — Results Summary

Summary Metrics (Test Set):
ResNet-50 (frozen conv backbone) | Test | Acc=0.9819 Prec=0.983


Training vs Validation Accuracy (ResNet-50)
Accuracy
Epoch
0.98
0.96
0.94
0.92
0.90
0.88
2
4
6
8
10
Train
Validation


Training vs Validation Loss (ResNet-50)
Train
Validation
Loss
0.35
0.30
0.25
0.20
0.15
0.10
2
4
6
8
10
Epoch


ROC Curve (Test Set)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC (AUC = 0.999)


Confusion Matrix (Test Set)
rotten
fresh
True label
249
6
5
347
rotten
fresh
Predicted label
300
250
200
150
100
50

Comparison Table

Model	Split	Accuracy	Precision	Recall	F1	AUC
Resnet-50 (frozen conv backbone)	Test	0.9819	0.983	0.9858	0.9844	0.9988